NEIL G. SENECAL

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EDUCATION

Brigham Young University
Ira A. Fulton College of Engineering and Technology September 2019 – Present
3.25 GPA | Expected Graduation: Apr 2027

Livermore High School
Green Engineering Academy August 2015 – June 2019
High School Diploma (*summa cum laude)*

EXPERIENCE

PERSONAL PROJECTS

Peltier-Based Water Reclamation System

- Designed and prototyped thermoelectric condensation system integrated into bathroom ventilator. **Awarded "Best of Show"** at school district science fair.

Electromechanical clock

- Designed and prototyped a smart clock with analog co-centric ring displays driven by a custom 12:1 gear train. Adapted an open-source Python library to drive an integrated LCD display. (More information)

Penny CPU Cooler

- Designed and manufactured a 3D-printed Raspberry Pi cooler using recycled pennies. (More information)

VroidPoser

- Developed a full-stack Python app for the recording and transmission of motion capture data via the OSC protocol. (More information)

Reactive Awning Control System

- Programmed C++ embedded system to autonomously trigger integrated actuators to adjust awning based on real-time weather sensor data.

DIY Alkaline Batteries

- Constructed and experimentally tested carbon-magnesium and copper-magnesium battery chemistries with various household electrolytes.

PROFESSIONAL AND VOLUNTEER

Computer Science Teacher's Assistant | *Brigham Young University* September 2024 – Present

- Mentor students in Python programming and debugging and contribute to curriculum development.

Shift Supervisor | *Harold B. Lee Library Cafe, BYU* August 2023 – September 2024

- Coordinated team operations and task prioritization. Conducted inventory audits and safety monitoring.

Missionary | *The Church of Jesus Christ of Latter-day Saints* March 2021 – March 2023

- Served two-year mission in New Jersey. Taught ESL classes in Korean and English. Developed cross-cultural communication and situational adaptability.

Primary Electrician | *FIRST Robotics Team #6619, Livermore High School* August 2018 – June 2019

- Designed electrical systems for competitive robotics platforms. Collaborated with cross-functional teams to integrate sensors, motors, and control systems.